

02 — Frontend Architecture

React/TypeScript architecture, state, routing and delivery

CampusFlow • Production Specification • v1.0 • October 2026

1. Frontend Stack

React + TypeScript + Vite.

Tailwind CSS.

Lucide React.

React Router.

TanStack Query.

React Hook Form + Zod.

2. Exact Folder Structure

```
src/  
  app/  
    router.tsx  
    providers.tsx  
  components/  
    ui/  
      layout/  
      notices/  
      events/  
      deadlines/  
  features/  
    auth/  
    dashboard/  
    notices/  
    events/  
    deadlines/  
    calendar/  
    admin/  
  pages/  
  hooks/  
  lib/  
    api.ts  
    auth.ts  
    validation.ts  
  types/  
  styles/
```

3. Implementation Rules

- Mobile-first.
- No inline styles.
- Use design tokens instead of scattered colors.
- Every page has appropriate loading, error and empty states.
- TanStack Query owns server state; do not duplicate it.
- API access goes through a shared client.
- All forms use validation.
- Prevent duplicate submissions.
- Use semantic HTML and visible focus states.

4. Routes

/login — public.

/dashboard, /notices, /deadlines, /events, /calendar, /notifications, /profile — authenticated.

/notices/:id and /events/:id — authenticated; server checks access.

/admin/* — role-protected.

5. State Architecture

Server state: TanStack Query.

Transient UI state: component state.

Auth/session: auth provider.

After mutations, invalidate only affected query keys.

6. Responsive Rules

Desktop: sidebar + header.

Tablet: collapsible sidebar.

Mobile: drawer or bottom navigation.

Tables become cards where needed.

Primary action remains visible.

Touch targets remain usable.

7. Three-Member Roadmap

Phase 0 Foundation: repo, stack, database, tokens, auth skeleton.

Phase 1 Core Student: login, dashboard, notices, deadlines, events, registration.

Phase 2 Admin: publishing, registrations, audit logs.

Phase 3 Hardening: security, tests, accessibility, responsive QA, monitoring.

Phase 4 Intelligence: AI summaries, ranking, reminders and semantic search.